

Relation Extraction

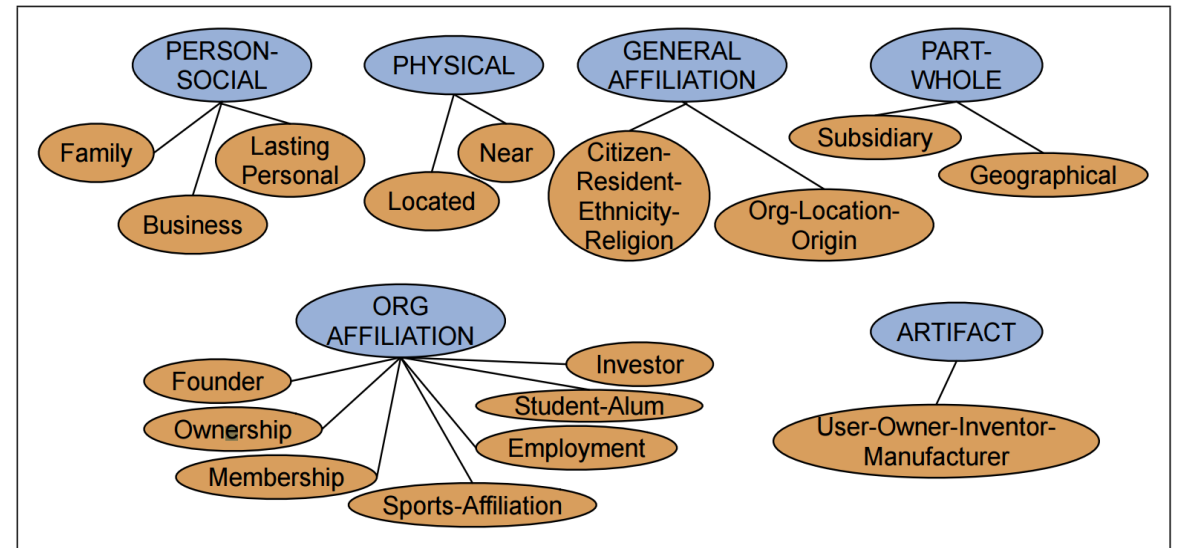
- Next on our list of tasks is to discern the relationships that exist among the detected entities

Citing high fuel prices, **United Airlines** said Friday it has increased fares by \$6 per round trip on flights to some cities also served by lowercost carriers. **American Airlines**, a unit of **AMR Corp.**, immediately matched the move, spokesman **Tim Wagner** said. **United**, a unit of **UAL Corp.**, said the increase took effect Thursday and applies to most routes where it competes against discount carriers, such as **Chicago** to **Dallas** and **Denver** to San Francisco.

- For instance, the text tells us that Tim Wagner is a spokesman for American Airlines, that United is a unit of UAL Corp., and that American Airlines is a unit of AMR

Relation Extraction

- Often the relations are mapped to a pre-defined ontology
 - This is an example from the ACE shared task:
- For instance the aforementioned relations are instances of the PART-WHOLE relation:
 - “United is a unit of UAL Corp.”
 - “American is a unit of AMR”



Ontologies can be Mapped to Model-Theoretic Semantics

Domain

United, UAL, American Airlines, AMR
Tim Wagner
Chicago, Dallas, Denver, and San Francisco

$$\mathcal{D} = \{a, b, c, d, e, f, g, h, i\}$$

$$a, b, c, d$$

$$e$$

$$f, g, h, i$$

Classes

United, UAL, American, and AMR are organizations
Tim Wagner is a person
Chicago, Dallas, Denver, and San Francisco are places

$$Org = \{a, b, c, d\}$$

$$Pers = \{e\}$$

$$Loc = \{f, g, h, i\}$$

Relations

United is a unit of UAL
American is a unit of AMR
Tim Wagner works for American Airlines
United serves Chicago, Dallas, Denver, and San Francisco

$$PartOf = \{\langle a, b \rangle, \langle c, d \rangle\}$$

$$OrgAff = \{\langle c, e \rangle\}$$

$$Serves = \{\langle a, f \rangle, \langle a, g \rangle, \langle a, h \rangle, \langle a, i \rangle\}$$

Relation Extraction

- There are many other ontologies of relations that have been defined
 - For example, UMLS, the Unified Medical Language System from the US National Library of Medicine has a network that defines 134 broad subject categories, such as:

Entity	Relation	Entity
Injury	disrupts	Physiological Function
Bodily Location	location-of	Biologic Function
Anatomical Structure	part-of	Organism
Pharmacologic Substance	causes	Pathological Function
Pharmacologic Substance	treats	Pathologic Function

Wikipedia Infoboxes

- Wikipedia also offers a range of relation types, drawn from the infoboxes
- For example, the Wikipedia infobox for Brad Pitt includes structured facts like *Occupation* = “Actor”/“Producer” or *Born* = “Shawnee, Oklahoma”
- These facts can be turned into relations like *born_in* or *occupied_as*
- These relations can complement relations extracted from text, or serve as features for them
- *Wikidata* (<https://www.wikidata.org>) is a knowledge base based on Wikipedia

Brad Pitt	
	
Pitt at the premiere of <i>Fury</i> in Washington, D.C. , October 2014	
Born	William Bradley Pitt December 18, 1963 (age 53) Shawnee, Oklahoma, U.S.
Occupation	Actor • producer
Years active	1987–present
Works	Filmography
Home town	Springfield, Missouri
Spouse(s)	Jennifer Aniston (m. 2000; div. 2005) Angelina Jolie (m. 2014; separated 2016)
Children	6
Relatives	Douglas Pitt (brother)

Using Patterns to Discover Relations

- The earliest and still common approach for relation extraction is the use of lexico-syntactic patterns

NP {, NP}* {,} (and or) other NP _H	temples, treasuries, and other important civic buildings
NP _H such as {NP,}* {(or and)} NP	red algae such as Gelidium
such NP _H as {NP,}* {(or and)} NP	such authors as Herrick, Goldsmith, and Shakespeare
NP _H {,} including {NP,}* {(or and)} NP	common-law countries , including Canada and England
NP _H {,} especially {NP}* {(or and)} NP	European countries , especially France, England, and Spain

Hearst, M. Automatic Acquisition of Hyponyms from Large Text Corpora, Proceedings of the Fourteenth International Conference on Computational Linguistics, 1992

Using Patterns to Discover Relations

- More modern approaches use additional features to define the patterns, such as named entity constraints
- For instance, if our goal is to answer questions about “who holds what office in which organization?”, we can use patterns like the following:

PER, POSITION of ORG:

George Marshall, Secretary of State of the United States

PER (named|appointed|chose|etc.) PER Prep? POSITION

Truman appointed Marshall Secretary of State

PER [be]? (named|appointed|etc.) Prep? ORG POSITION

George Marshall was named US Secretary of State

Using Patterns to Discover Relations

- More accurate patterns still are ones that use syntactic information
- For instance, symmetric patterns can be used to discover synonymy or relatedness between entities

from **X** to **Y**

X and **Y**

X or **Y**

neither **X** nor **Y**

X as well as **Y**

Using Patterns to Discover Relations

- Syntactic information can filter a lot of noise
- What are the *Xs* and *Ys* in these cases:

- “when they go to Austria, they like walking in the woods **as well as** skiing”
- “apricots **and** other vegetables **and** fruit”
- “Sandy is a Republican **and** proud of it”

from **X** to **Y**

X and **Y**

X or **Y**

neither **X** nor **Y**

X as well as **Y**

Neural Information Extraction

- The simplest approach is to assume relations come from a closed set and to predict the relation given a sentence, where the subject and object are replaced with their NER tags
 - Assumes a binary relation

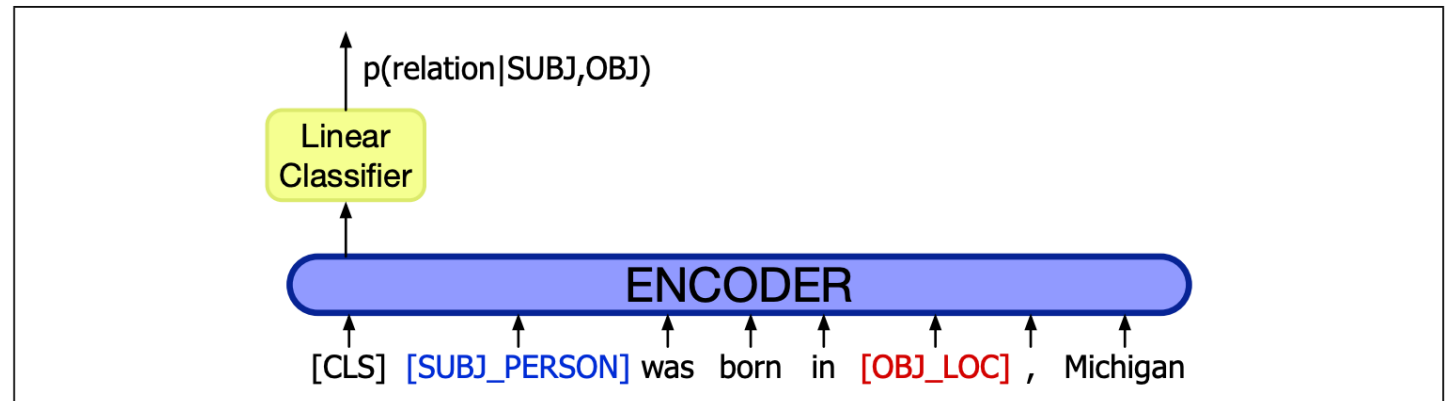


Figure 21.7 Relation extraction as a linear layer on top of an encoder (in this case BERT), with the subject and object entities replaced in the input by their NER tags (Zhang et al. 2017, Joshi et al. 2020).

Evaluation of Relation Extraction

- Semi-supervised methods are much more difficult to evaluate than supervised methods
 - They extract **new** relations from the web or a large text
 - As methods use very large amounts of text, it is impossible to test them in a sand box with a small pre-annotated gold standard
- Evaluation is therefore done by:
 - Computing precision
 - If the system can rank its output, we can measure precision for different output sizes (e.g., precision for 100 relations, 1000 relations etc.)
 - Extrinsic evaluation by testing how well these relations help, say, question answering

Poon and Domingos, *Unsupervised Semantic Parsing*,
EMNLP 2009